

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary End Semester Examination – Summer 2022**

**Course: B. Tech.**

**Branch : Computer Engineering**

**Semester : VII**

**Subject Code & Name: Fundamental of Digital Image Processing (BTCOE702C)**

**Max Marks: 60**

**Date:18/08/2022**

**Duration: 3.45 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)      Marks

**Q. 1 Solve Any Two of the following.**

- |                                                                                                                                                                         |                   |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| <b>A)</b> Define Image and describe the elements that helps to compose an image? What are the different types of images?                                                | <b>Understand</b> | <b>[6]</b> |
| <b>B)</b> Explain the fundamental steps performed in Digital Image Processing with suitable examples. Also, Explain in detail the concept of sampling and quantization. | <b>Understand</b> | <b>[6]</b> |
| <b>C)</b> How pixels are related? Explain various types of correlation between pixels. Explain the pixel connectivity.                                                  | <b>Understand</b> | <b>[6]</b> |

**Q.2 Solve Any Two of the following.**

- |                                                                                                                    |                   |            |
|--------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| <b>A)</b> What is image enhancement? Enlist and explain are the frequency domain techniques for image enhancement. | <b>Understand</b> | <b>[6]</b> |
| <b>B)</b> What is image contrast? Explain in brief point processing technique.                                     | <b>Understand</b> | <b>[6]</b> |
| <b>C)</b> What is image smoothing? Identify various techniques used for image smoothness.                          | <b>Apply</b>      | <b>[6]</b> |

**Q. 3 Solve the following questions.**

- |                                                                                                      |                   |            |
|------------------------------------------------------------------------------------------------------|-------------------|------------|
| <b>A)</b> What is image compression? Explain types of redundancies present in the image.             | <b>Understand</b> | <b>[6]</b> |
| <b>B)</b> Explain in details basic steps in image compression? Enlist the need of image compression? | <b>Understand</b> | <b>[6]</b> |

**Q.4 Solve Any Two of the following.**

- |                                                                                                                                                               |                   |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|
| <b>A)</b> What is image segmentation? Describe the following Image Segmentation Technique.<br>(a) Threshold Based Segmentation<br>(b) Edge Based Segmentation | <b>Understand</b> | <b>[6]</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|

**B)** Explain the following image segmentation techniques.

- (a) Region-Based Segmentation
- (b) Clustering Based Segmentation

**Understand [6]**

**C)** How to detect discontinuity in the image? Which are the types of discontinuities in the image?

**Remember [6]**

**Q. 5 Solve Any One of the following.**

**A)** Identify and describe in details any six applications of image processing in satellite.

**Apply [12]**

**B)** Explain Medical Image processing in detail.

**Understand [12]**

**\*\*\* End \*\*\***